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Department of Electronics & Communication Engineering | GATE Preparation Library

Communication Systems — Essential Formula Guide

AM/FM Equations, Carson's Rule, AWGN Channel Capacity, PCM SQNR & Digital Modulation BER

GATE Section: Section 7 • **Marks:** 9 - 12 Marks • **Core Focus:** Spectral Efficiency, BER & Information Theory

DIGITAL MODULATION & BIT ERROR PROBABILITIES
- **BPSK & QPSK:** Pe = Q(sqrt(2 * Eb / N0)). (QPSK gives double data rate for identical bandwidth and BER!).
- **Coherent BFSK:** Pe = Q(sqrt(Eb / N0)) (Requires 3 dB more power than BPSK).
- **Shannon-Hartley Capacity:** C = B * log2(1 + S / (N0 * B)) bits/second. As B -> inf, C_max = 1.442 * (S / N0).

SECTION: ANALOG MODULATION (AM, FM, PM)

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Standard AM (DSB-FC)	**s(t) = Ac * [1 + mu * cos(2*pi*fm*t)] * cos(2*pi*fc*t)** **P_total = P_c * (1 + mu^2 / 2)**	mu = Am / Ac (modulation index) Bandwidth = 2 * fm	Efficiency: eta = mu^2 / (2 + mu^2). Max efficiency for sinusoidal modulation = 33.3% (at mu = 1).
DSB-SC & SSB-SC	**DSB-SC: s(t) = Ac * m(t) * cos(2*pi*fc*t), BW = 2*fm** **SSB-SC: s(t) = Ac*m(t)*cos(wc*t) +- Ac*m_hat(t)*sin(wc*t), BW = fm**	m_hat(t) = Hilbert transform of m(t) 100% power efficiency	SSB-SC has smallest transmission bandwidth of all analog systems.
Carson's Rule for FM	**BW = 2 * (delta_f + f_m) = 2 * f_m * (beta + 1)**	delta_f = kf * Am (frequency deviation) beta = delta_f / fm (modulation index)	Narrowband FM (beta << 1): BW ~ 2*fm. Wideband FM (beta >> 1): BW ~ 2*delta_f.
Superheterodyne Receiver	**f_LO = f_s + f_IF (high-side tuning)** **f_image = f_s + 2 * f_IF**	f_IF = 455 kHz (AM), 10.7 MHz (FM)	Image Rejection Ratio: IRR = sqrt(1 + Q^2 * rho^2) where rho = (f_img/f_s) - (f_s/f_img).

SECTION: INFORMATION THEORY, ENTROPY & CHANNEL CAPACITY

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Entropy of Source	**H(X) = -sum_{i=1}^M [P(xi) * log2 P(xi)] bits/symbol**	H_max = log2(M) for equiprobable symbols	Information rate: R = r * H(X) bits/second (r = symbol generation rate).
Binary Entropy Function	**H(p) = -p * log2(p) - (1 - p) * log2(1 - p)**	p = probability of 1, (1-p) = prob of 0	Maximum at p = 0.5 where H(p) = 1 bit/symbol.
Channel Capacity (AWGN)	**C = B * log2(1 + S / (N0 * B)) bps**	B = channel bandwidth, S = signal power N0/2 = noise power spectral density	If SNR -> inf, capacity grows linearly with bandwidth. If B -> inf, C -> 1.442 * (S / N0).
Binary Symmetric Channel	**C = 1 - H(p) = 1 + p*log2(p) + (1-p)*log2(1-p)**	p = transition / crossover probability	If p = 0.5, C = 0 (channel is completely useless). If p = 0 or 1, C = 1 bit/channel use.

SECTION: DIGITAL BASEBAND (PCM) & PASSBAND MODULATION

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
PCM Quantization & SNR	**Step size: delta = (V_max - V_min) / 2^n** **N_q = delta^2 / 12** **(SQNR)_dB = 6.02 * n + 1.76 dB**	n = number of bits per sample Bit rate: R_b = n * f_s >= 2 * n * f_m	Each additional bit adds ~6 dB to the signal-to-quantization noise ratio!
Delta Modulation Slope Limit	**A_max <= (delta * f_s) / (2 * pi * f_m)**	delta = step size, f_s = sampling frequency	Slope overload distortion occurs when |dm(t)/dt| > delta * f_s. Avoided by increasing delta or f_s.
Matched Filter Peak SNR	**h(t) = s(T - t)** **(SNR_out)_max = 2 * E / N0**	E = energy of received signal s(t) N0/2 = white noise PSD	Output SNR depends ONLY on signal energy E, completely independent of waveform shape!
BPSK vs QPSK vs BFSK	**BPSK: Pe = Q(sqrt(2*Eb/N0)), BW = 2*Rb** **QPSK: Pe = Q(sqrt(2*Eb/N0)), BW = Rb** **BFSK: Pe = Q(sqrt(Eb/N0)), BW = 2*Rb**	Eb = energy per bit, Rb = bit rate Q(x) = (1/sqrt(2*pi)) * integral_x^inf e^(-u^2/2) du	QPSK transmits 2 bits/symbol, achieving half the bandwidth of BPSK with identical BER performance.